

CARBON MARKETS CCUS AND DIRECT AIR CAPTURE

CARBON-UNIT IDENTITY FORK → MARKET ARCHITECTURE MATRIX → CLAIM-TYPE LADDER → INTEGRITY GATE → CREDIT LIFECYCLE RAIL → ACCOUNTING-RISK FIREWALL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 CARBON-UNIT IDENTITY FORK

CARBON-UNIT IDENTITY FORK PERMISSION UNIT INSIDE A COMPLIANCE CONSTRAINT VERIFIED OUTCOME AGAINST AN APPROVED BASELINE ALLOCATION OR AUCTION VERSUS METHODOLOGY AND ISSUANCE SURRENDER OR RETIREMENT UNDER THE APPLICABLE RULE

TRADABILITY DOES NOT ERASE DIFFERENT UNIT ORIGINS

ALLOWANCE → permission unit inside a compliance constraint

CREDIT → verified outcome against an approved baseline

ORIGIN → allocation or auction versus methodology and issuance

USE → surrender or retirement under the applicable rule

SYSTEM / LEVEL / DEFINITION

- ALLOWANCE → permission unit inside a compliance constraint
- CREDIT → verified outcome against an approved baseline

STRUCTURE-FUNCTION MECHANISM

- ORIGIN → allocation or auction versus methodology and issuance
- USE → surrender or retirement under the applicable rule

SCALE / EVIDENCE LIMIT

- TRAP → tradability does not erase different unit origins
- ALLOWANCE → permission unit inside a compliance constraint

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → tradability does not erase different unit origins.

01

STAGE 01 MARKET ARCHITECTURE MATRIX

MARKET ARCHITECTURE MATRIX CAP AND TRADE ABSOLUTE COVERED-EMISSIONS CEILING BASELINE AND CREDIT PERFORMANCE AGAINST TARGET OR BASELINE LEGAL OBLIGATION AND SURRENDER REQUIREMENT NON-MANDATED PURCHASE OR CLAIM

WHAT CONSTRAINS EMISSIONS AND WHO MUST ACT

ECOLOGICAL UNIT / PARAMETER

CAP AND TRADE → absolute covered-emissions ceiling

BASELINE AND CREDIT → performance against target or baseline

DECISIVE DISTINCTION

COMPLIANCE → legal obligation and surrender requirement

VOLUNTARY → non-mandated purchase or claim

SCALE / STATUS / EXCEPTION

QUESTION → what constrains emissions and who must act

CAP AND TRADE → absolute covered-emissions ceiling

ECOLOGICAL UNIT / PARAMETER

CAP AND TRADE → absolute covered-emissions ceiling

BASELINE AND CREDIT → performance against target or baseline

DECISIVE DISTINCTION

COMPLIANCE → legal obligation and surrender requirement

VOLUNTARY → non-mandated purchase or claim

SCALE / STATUS / EXCEPTION

QUESTION → what constrains emissions and who must act

CAP AND TRADE → absolute covered-emissions ceiling

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUESTION → what constrains emissions and who must act.

02

STAGE 02 CLAIM-TYPE LADDER

CLAIM-TYPE LADDER COUNTERFACTUAL SOURCE DOES NOT ARISE EXISTING SOURCE EMITS LESS CARBON LEAVES AMBIENT ATMOSPHERE DURATION AND REVERSAL RISK DETERMINE DURABILITY NO MERGER EACH SUPPORTS A DIFFERENT CLIMATE CLAIM

AVOIDANCE → counterfactual source does not arise

REDUCTION → existing source emits less

REMOVAL → carbon leaves ambient atmosphere

STORAGE → duration and reversal risk determine durability

SYSTEM / LEVEL / DEFINITION

- AVOIDANCE → counterfactual source does not arise
- REDUCTION → existing source emits less

STRUCTURE-FUNCTION MECHANISM

- REMOVAL → carbon leaves ambient atmosphere
- STORAGE → duration and reversal risk determine durability

SCALE / EVIDENCE LIMIT

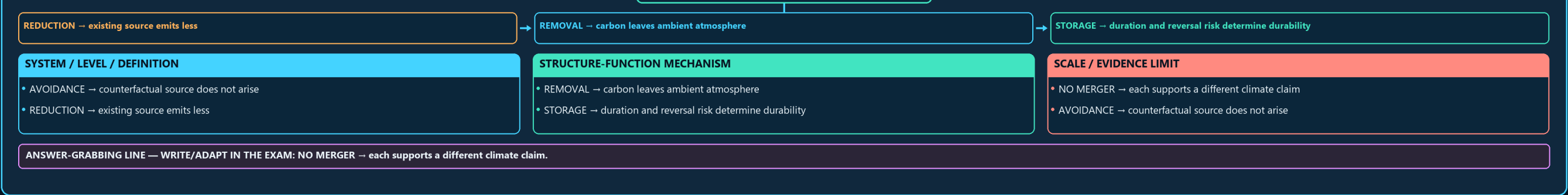
- NO MERGER → each supports a different climate claim
- AVOIDANCE → counterfactual source does not arise

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO MERGER → each supports a different climate claim.

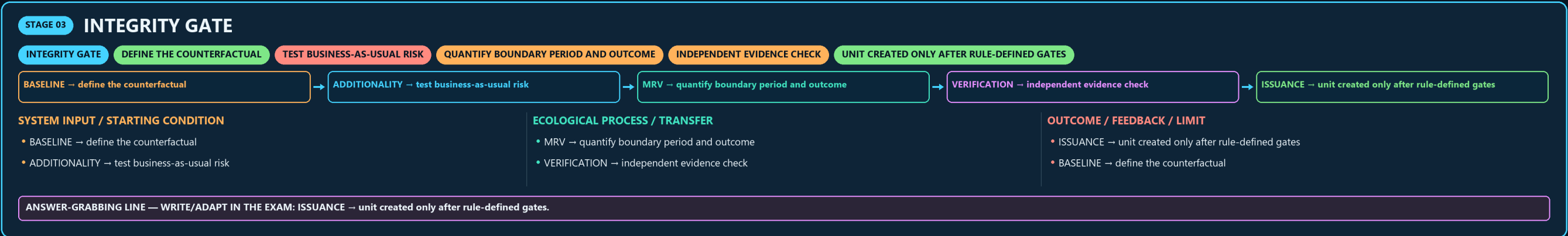
03

STAGE 03 INTEGRITY GATE

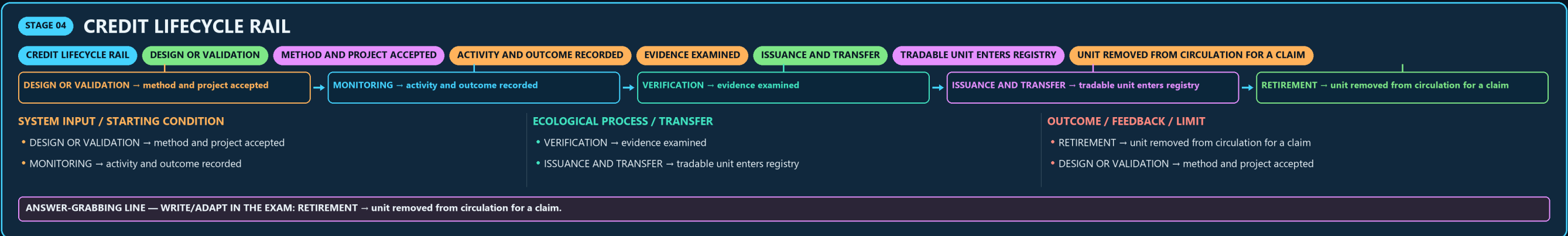
INTEGRITY GATE DEFINE THE COUNTERFACTUAL TEST BUSINESS-AS-USUAL RISK QUANTIFY BOUNDARY PERIOD AND OUTCOME INDEPENDENT EVIDENCE CHECK UNIT CREATED ONLY AFTER RULE-DEFINED GATES



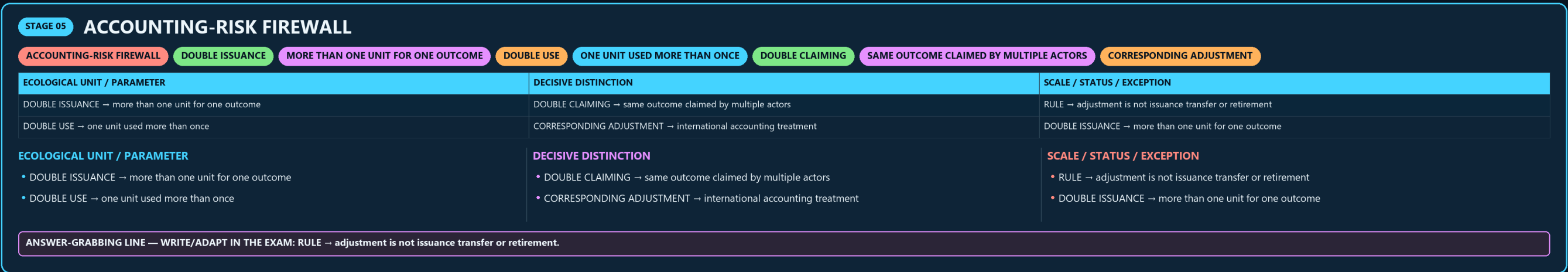
03



04

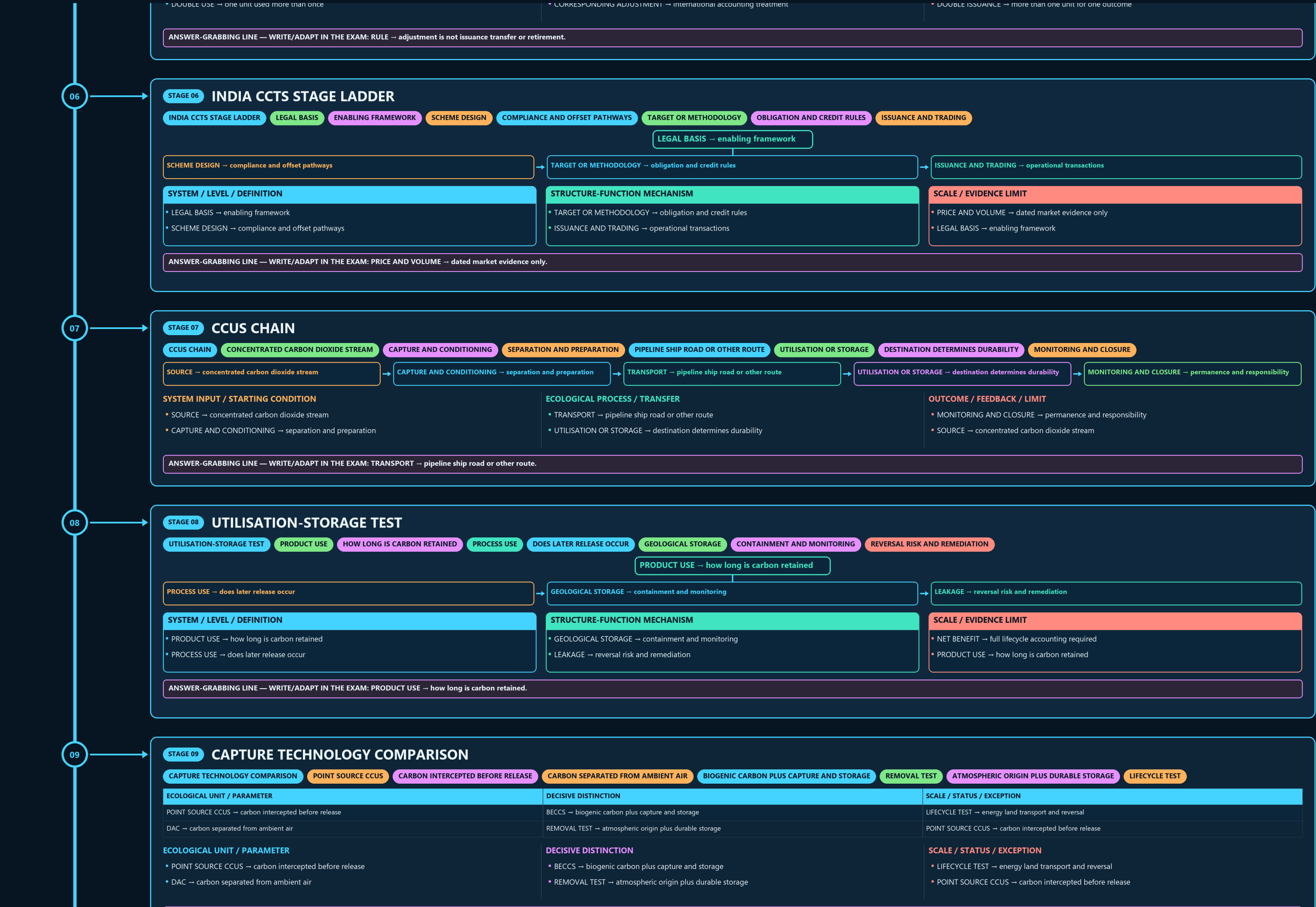


05



06





09

STAGE 09 CAPTURE TECHNOLOGY COMPARISON

CAPTURE TECHNOLOGY COMPARISON

POINT SOURCE CCUS

CARBON INTERCEPTED BEFORE RELEASE

CARBON SEPARATED FROM AMBIENT AIR

BIOGENIC CARBON PLUS CAPTURE AND STORAGE

REMOVAL TEST

ATMOSPHERIC ORIGIN PLUS DURABLE STORAGE

LIFECYCLE TEST

ECOLOGICAL UNIT / PARAMETER

POINT SOURCE CCUS → carbon intercepted before release

DAC → carbon separated from ambient air

DECISIVE DISTINCTION

BECCS → biogenic carbon plus capture and storage

REMOVAL TEST → atmospheric origin plus durable storage

SCALE / STATUS / EXCEPTION

LIFECYCLE TEST → energy land transport and reversal

POINT SOURCE CCUS → carbon intercepted before release

ECOLOGICAL UNIT / PARAMETER

• POINT SOURCE CCUS → carbon intercepted before release

• DAC → carbon separated from ambient air

DECISIVE DISTINCTION

• BECCS → biogenic carbon plus capture and storage

• REMOVAL TEST → atmospheric origin plus durable storage

SCALE / STATUS / EXCEPTION

• LIFECYCLE TEST → energy land transport and reversal

• POINT SOURCE CCUS → carbon intercepted before release

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIFECYCLE TEST → energy land transport and reversal.

10

STAGE 10 NET-ZERO MITIGATION HIERARCHY

NET-ZERO MITIGATION HIERARCHY

PREVENT UNNECESSARY EMISSIONS

DECARBONISE EXISTING ACTIVITIES

ADDRESS HARD-TO-ABATE POINT SOURCES

COUNTERBALANCE GENUINELY RESIDUAL EMISSIONS

NET BALANCE NEEDS ONE BOUNDARY AND PERIOD

AVOID → prevent unnecessary emissions

REDUCE → decarbonise existing activities

CAPTURE → address hard-to-abate point sources

REMOVE → counterbalance genuinely residual emissions

SYSTEM / LEVEL / DEFINITION

• AVOID → prevent unnecessary emissions

• REDUCE → decarbonise existing activities

STRUCTURE-FUNCTION MECHANISM

• CAPTURE → address hard-to-abate point sources

• REMOVE → counterbalance genuinely residual emissions

SCALE / EVIDENCE LIMIT

• VERIFY → net balance needs one boundary and period

• AVOID → prevent unnecessary emissions

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → net balance needs one boundary and period.

11

STAGE 11 CARBON INSTRUMENTS ANSWER SPINE

CARBON INSTRUMENTS ANSWER SPINE

UNIT MARKET AND CLAIM TYPE

BASELINE MRV ISSUANCE TRANSFER RETIREMENT

CAPTURE TRANSPORT UTILISATION OR STORAGE

PERMANENCE LEAKAGE LIFECYCLE AND DOUBLE COUNTING

CURRENT RULE PRICE VOLUME COST AND CAPACITY SOURCE

CLASSIFY → unit market and claim type

TRACE → baseline MRV issuance transfer retirement

FOLLOW → capture transport utilisation or storage

TEST → permanence leakage lifecycle and double counting

AUDIT → current rule price volume cost and capacity source

SYSTEM / LEVEL / DEFINITION

• CLASSIFY → unit market and claim type

• TRACE → baseline MRV issuance transfer retirement

STRUCTURE-FUNCTION MECHANISM

• FOLLOW → capture transport utilisation or storage

• TEST → permanence leakage lifecycle and double counting

SCALE / EVIDENCE LIMIT

• AUDIT → current rule price volume cost and capacity source

• CLASSIFY → unit market and claim type

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AUDIT → current rule price volume cost and capacity source.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

DAC'S FUNDAMENTAL TECHNICAL CHALLENGE IS THERMODYNAMIC

ATMOSPHERIC CO₂ CONCENTRATION

POLICY IMPLICATION

BECAUSE DAC DELIVERS GENUINE ATMOSPHERIC REMOVAL (UNLIKE

GLOBAL CCUS AND DAC DEPLOYMENT CAPACITY REMAINS SMALL RELATIVE

THE INDIAN CARBON MARKET'S SPECIFIC SECTORAL COVERAGE, TARGET-SETTING METHODOLOGY

MRV-SYSTEM ROBUSTNESS FOR BOTH INDIA'S DOMESTIC CARBON MARKET AND

OPTIONAL ECOLOGICAL PROCESS DEPTH

• CAUTION: DAC's fundamental technical challenge is thermodynamic: atmospheric CO₂ concentration

• CAUTION: Policy implication: because DAC delivers genuine atmospheric removal (unlike

• CAUTION: Global CCUS and DAC deployment capacity remains small relative to the scale IPCC

• CAUTION: The Indian Carbon Market's specific sectoral coverage, target-setting methodology and

SCALE, PARAMETER OR STATUS LIMIT

• CAUTION: MRV-system robustness for both India's domestic carbon market and international

• India's carbon market has a two-track design: a compliance mechanism (building on PAT,

• Standard fossil-source CCUS is an emissions-avoidance technology; genuine atmospheric

• DAC's higher cost/energy-intensity relative to CCUS stems from the much lower CO₂

COMPARATIVE RESTORATION NUANCE

• Net-zero frameworks require genuine removal technologies (DAC/BECCS/nature-based sinks)

• NOT: All CCUS applications, including standard fossil-power-plant CCUS, deliver negative

• NOT: India's carbon market is a single-track, purely compliance-based system. - It has a

• NOT: DAC is currently deployed at a scale comparable to global CCUS capacity. - DAC remains

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.